

IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

EXAMINATIONS 2024

BEng Honours Degree in Computing Part I
MEng Honours Degrees in Computing Part I
for Internal Students of the Imperial College of Science, Technology and Medicine

*This paper is also taken for the relevant examinations for the
Associateship of the City and Guilds of London Institute*

PAPER COMP40018A

DISCRETE MATHEMATICS

Friday 17th May 2024, 14:00
Duration: 40 minutes

Answer ALL ONE question

This is a CLOSED BOOK exam

Paper contains 1 question
Calculators not required

1 a Let $A = \{a, \{a\}, \{b\}\}$ and $B = \{\{a\}, b\}$. Determine the *eight* sets $A \cap B$, $A \cup B$, $A \triangle B$, $\varnothing B$, $\varnothing B \setminus B$, $A \cap \varnothing B$, $A \times B$, and $(A \times B) \cap (B \times A)$.

b Show that $\overline{R \cup S}^{-1} = \overline{R^{-1}} \cap \overline{S}$.

You do not need to give a formal proof, and can use the abbreviated notation for statements. Write clearly when a step follows by definition, or when by a logical reasoning, or when information from the notes is used.

c Let $\mathcal{P} \triangleq \{2, 3, 5, 7, 11, 13, \dots\}$ be the set of prime numbers. We define:

$$\begin{aligned} V_n &\triangleq \{p \in \mathcal{P} \mid \exists m \in \mathbb{N} (p \times m = n)\} \\ \sqsubseteq &\triangleq \{\langle n, m \rangle \in \mathbb{N}^2 \mid V_n \subseteq V_m\} \end{aligned}$$

Does $\sqsubseteq$ have minimal, least, maximal, or greatest elements? If yes, is it unique? If no, argue why. Give examples, and show why each satisfies the criterion.

d Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijective functions. Prove that $g \circ f$ is a bijective function as well.

e Prove that if A is countable, then A^2 is as well. (State clearly any result from the course that you may use in your proof.)

The five parts carry equal marks.